Research on Tennis Match Strategies Based on Machine Learning and Markov Chain Modeling

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Abstract. This paper mainly studies the influence of athletes' performance on the results of sports competitions, and uses advanced data analysis and machine learning technology to predict the players' scoring patterns in tennis matches. Firstly, the Markov chain model is used to capture match points and game flow. In order to quantify player performance, the concept of momentum is introduced. The random forest model based on factor analysis optimization studies the factors affecting player performance. Secondly, the random forest model optimized by firefly algorithm is used to select the factors that may affect the competition, identify the important factors and data that affect the fluctuation of the competition, and realize the prediction. This study provides an advanced perspective on sports analytics, providing actionable insights for athletes and coaches to refine their strategies and gain a competitive edge, while also providing lessons for the development of sports analytics and data-driven decision making.

Keywords: Machine Learning, Data Analysis, Tennis Strategy, Random Forest Algorithm.

1. Introduction

Factors affecting the game, often valued by athletes and coaches, to Figure out the rea-sons for its generation and the mechanism affecting the game, is conducive to the athletes as well as the coach's pre-match preparation work and on-site guidance and adjustments to be able to do targeted, to ensure that the athletes play, and to further improve the athletes' athletic performance in the game [1].

This paper collects data from the Wimbledon men's singles tournament, which contains detailed information such as the names and scores of the players in the tournament, and through in-depth analysis of the valid data, machine learning and predictive modeling methods are implemented to study the strategy of the tennis tournament. A mathematical model that can capture the flow of player scoring in a tennis match and determine which player performs better at a given point in time is developed along with visualization tools to show the progress of the match. Fully consider the influencing factor in the model, i.e., the player who serves usually has a higher probability of winning a point/set in a tennis match. Based on this, use the available data to further enable prediction of turning points in the game and identify key factors.

2. Process Capture and Performance Quantification

2.1. Data preprocessing

First of all, the data processing work mainly includes the following.

1. label encoding. There are many categorical variables in the data, presented as strings. In order to make it easy for the model to accept this type of data, it needs to be preprocessed by means of label encoding. The last three columns of the tennis match data are encoded and for each column, a unique integer value is assigned to each different label.
2. Removing outliers. Outliers may negatively affect the model, use the triple standard method to check for and deal with outliers. Ensure data accuracy and consistency.
3. Filling in missing values. The dataset is examined and it is found that there are missing values in the data table, so the missing values need to be interpolated.

All manuscripts must be in English, also the table It is worth mentioning that in the data preprocessing, we used an unconventional BP neural network algorithm to accomplish the interpolation of missing

values. The com-parison of interpolation accuracy with mean value interpolation is shown in Fig.1 [2].

In Fig.1, the scatter is the predicted value of the BP neural network, the lake green dash line is the actual value, and the red straight line is the mean predicted value, which shows that the interpolation accuracy of our BP neural network algorithm is much higher than that of the commonly used mean interpolation method.

2.2. Capturing Match Points and Game Flow

The Markov chain model can handle transitions between directly observable states without involving hidden states. In the problem, each point in the match, each game, each set, and the serving side can be considered as states within the Markov chain, and transitions between these states are based on the current state rather than the historical sequence, adhering to the Markov property (Memory lessness) [3]. The model construction process is as follows:

1. Define the state space: In a tennis match, the state space encompasses all possible match score states (combinations of game and set scores) along with the current server. For example, a state could be "Player 1 has won 3 games, Player 2 has won 2 games, Player 1 has won 1 set, Player 2 has not won a set, and it is currently Player 1's serve."
2. Transition Probabilities: Based on match data, it is possible to calculate the transition probabilities from one state to another. For example, for the current state "3-2,1-0, Player 1 serving," the transition probability to the next state "4-2,1-0Player 2 serving" can be computed.
3. Constructing the Transition Matrix: Organize all the state transition probabilities into a matrix known as the transition matrix. Each row of the matrix represents a specific starting state, each column represents a specific destination state, and the elements in the matrix represent the transition probabilities. In this model, the state transition probabilities are determined based on the frequency of occurrence of score states, taking into account set scores and game scores, without considering individual point scores such as 15-0, 30-0, 40-0.

Utilizing the constructed Markov chain model, we conduct analysis to predict potential match outcomes, calculate probabilities of reaching specific states, and examine how a player's winning probability changes, especially concerning the serving side's situation.

After importing the data, we obtained the change in winning probabilities for both players throughout the entire match, as shown in Fig.2.

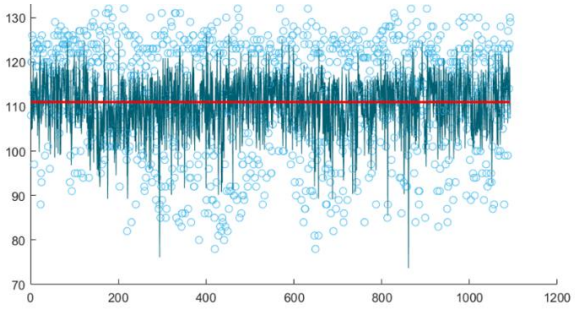


Figure.1 Hitting speed interpolation accuracy comparison

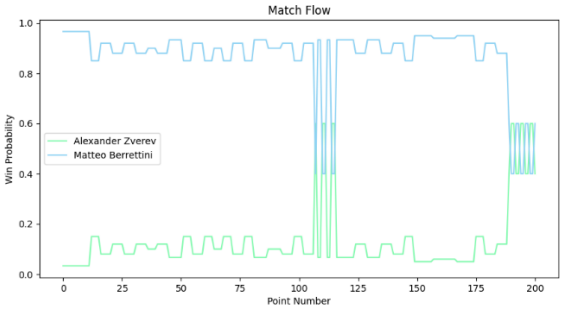


Figure.2 Process capture chart

2.3. Assessment of the “Momentum”

2.3.1 Performance Score Evaluation Model Based on FA-RF

Due to the large number of data types in the actual collection, it is necessary to reduce the dimension of features for more representative classification. In this paper, factor analysis is used to datum the interdependencies within the continuous attributes, grouping a large number of complex relationships into a small number of composite factors. The original system is represented as a new

combination of factors by analyzing the effect of variance contribution among the variables to reproduce the internal connections [4]. The model is constructed as follows:

$$\begin{cases} x_1 = a_{11}F_1 + a_{12}F_2 + \cdots + a_{1m}F_m + e_1 \\ x_2 = a_{21}F_1 + a_{22}F_2 + \cdots + a_{2m}F_m + e_2 \\ \cdots\cdots \\ x_p = a_pF_1 + a_{|2}F_2 + \cdots + a_{pm}F_m + e_p \end{cases} \tag{1}$$

In the above equation, $X = \left(x_1, x_2, \ldots, x_p\right)^T$ is observable random vector, i.e., they are the continuous attributes given in the question. And it satisfies $E(X) = 0, COV(X) = \sum$.

$F = (F_1, F_2, \ldots, F_m)^T (m < p)$ is the common factor, which is an unmeasurable random vector. And satisfy $E(F) = 0, Cov(F) = 1$, the vectors are independent of each other.

$\varepsilon = \left(e_1, e_2, \ldots, e_p\right)^T$ and F are independent of each other. and satisfies $E(e) = 0$, the e's are independent of each other.

Its matrix form is:

$$X = AF + \varepsilon \tag{2}$$

In this Eq, $A = \left(a_{ij}\right)_{p*m}$ is called the factor loading matrix.

Therefore, the factor score model can be written as:

$$\begin{cases} F_1 = \omega_{11}x_1 + \omega_{12}x_2 + \cdots + \omega_{1p}x_p \\ F_2 = \omega_{21}x_1 + \omega_{22}x_2 + \cdots + \omega_{2p}x_p \\ \cdots\cdots \\ F_m = \omega_{m1}x_1 + \omega_{m2}x_2 + \cdots + \omega_{mp}x_p \end{cases} \tag{3}$$

KMO and Bartlett's tests were performed on the data, and after a series of filters, the following features were selected in order to capture changes in athlete performance over time in the flow of the game. The features are shown in Table 1:

Table 1. Features

Feature	Calculation Method
points won difference	p1_points_won − p2_points_won
aces	statistical value
winners	statistical value
unforced errors	statistical value
distance run	statistical value, the distance run by the athlete during the round
break points won	statistical value
break points missed	statistical value
net points won	statistical value
server	statistical value

In assessing momentum changes in matches, we consider factors including points won difference, aces, winners, unforced errors, distance run, break points won, break point missed, net points won, and server advantage. We employed a random forest model to calculate the contribution of each factor to the momentum and computed the momentum score for each point by aggregating the weighted scores of these factors. The cumulative value of the momentum score reflects the changes in momentum throughout the match and can be visually represented using a line graph, providing an intuitive depiction of momentum fluctuations between the two players. This approach allows us to quantify momentum shifts during the match and provides a basis for strategic adjustments for coaches and players [5].

2.3.2 Visualization of Momentum

To showcase the game flow and momentum shifts, we utilize visualizations such as line charts and heatmaps. Line charts effectively represent the momentum scores over time, highlighting fluctuations in advantage between players. Heatmaps illustrate the intensity and frequency of winning points,

errors, and other key events in different segments of the game, providing a deeper insight into strategic patterns and pivotal moments. These visualizations not only allow for detailed analysis but also offer an intuitive grasp of the game dynamics.

Based on random forest analysis, determining the weights of various factors, the preliminary momentum calculation formula obtained is as follows:

$$M_t = \left(p_t + 2 \cdot a_t + 2 \cdot w_t - 2 \cdot u_t + n_t + 3 \cdot b_{w_t} - 2 \cdot b_{m_t} + \frac{d_t}{100}\right) \tag{4}$$

We conducted a study on the correlation between serving and winning, and plotted a confusion matrix as shown in Fig.3:

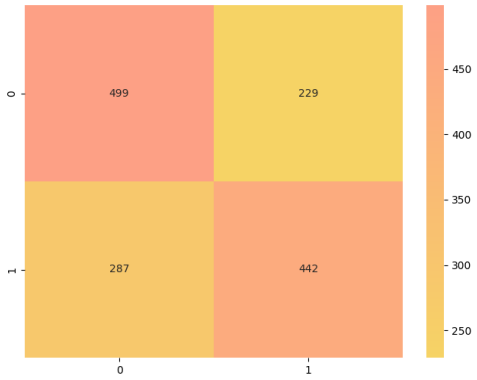


Figure.3 Confusion matrix

In theory, when serving, the overall score is expected to increase by 20% over the base score. That means when considering factors related to a high serving success rate, in the momentum score formula, a multiplication factor would be applied. The modified formula is as follows:

$$M_t = \left(p_t + 2 \cdot a_t + 2 \cdot w_t - 2 \cdot u_t + n_t + 3 \cdot b_{w_t} - 2 \cdot b_{m_t} + \frac{d_t}{100} + \delta\right) \tag{5}$$

$$\delta = \begin{cases} 0.2 & If\ the\ serving\ side\ wins\ the\ point. \\ 0 & If\ the\ serving\ side\ does\ not\ win\ the\ point. \end{cases} \tag{6}$$

After visualizing the scores obtained from this formula, we generated a line chart depicting Player 1's momentum changes over time, as shown in Fig.4. Additionally, we obtained a comparative chart of momentum for different players, as shown in Fig.5.

Based on the images, it becomes apparent that the momentum scores calculated using the initial formula exhibit excessive overall fluctuations. As the score gradually increases, the momentum becomes unmanageable. Therefore, we need to consider incorporating the influence of time to restrict momentum changes and judiciously account for the impact of negative factors on overall momentum. We also observed that winning streaks have a significant effect on momentum. Thus, we need to provide additional scoring for continuous scoring situations to make our model more closely resemble real-world scenarios.

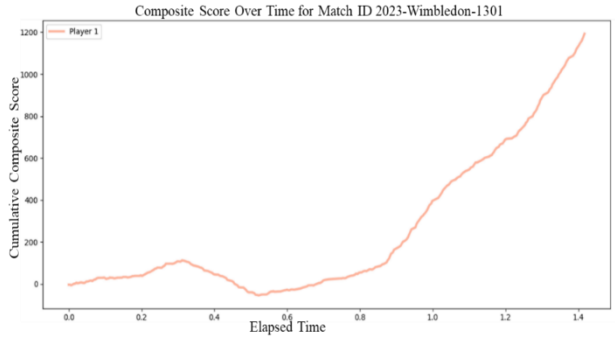


Figure.4 Player 1's momentum score over time line charts

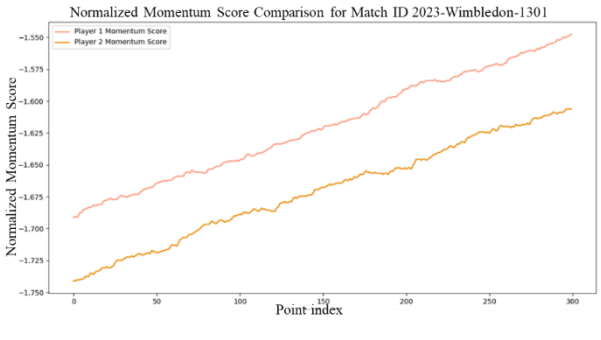


Figure.5 Comparison of momentum score line charts for both players

After implementing feature standardization constraints and considering the factors of time and winning streaks, we obtained the new momentum score formula as shown in formula (6):

$$M_t = \begin{bmatrix} \left(\frac{t_d}{166}\right) + 0.5 \cdot p_t + 0.1 \cdot n_t + a_t + 0.5 \cdot w_t \\ -0.3 \cdot u_t + 0.3 \cdot (1.1^{n_a} - 1) + 0.36 b_{w_t} - 0.24 b_{m_t} + \left(\frac{d}{400}\right) + \delta \end{bmatrix} \tag{7}$$

$$\delta = \begin{cases} 0.01 & If\ the\ serving\ side\ wins\ the\ point. \\ 0 & If\ the\ serving\ side\ does\ not\ win\ the\ point. \end{cases} \tag{8}$$

Based on the latest formula, we can obtain new momentum scores and visualize them. The changes in momentum scores for both players throughout the entire match and at specific times are shown in Fig.6 and Fig.7, respectively.

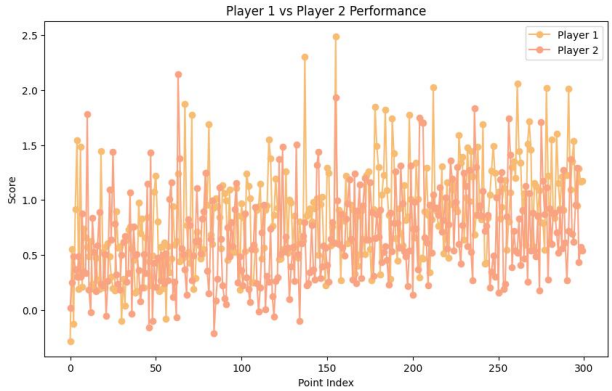


Figure.6 Comparison line chart of momentum for both players throughout the entire match

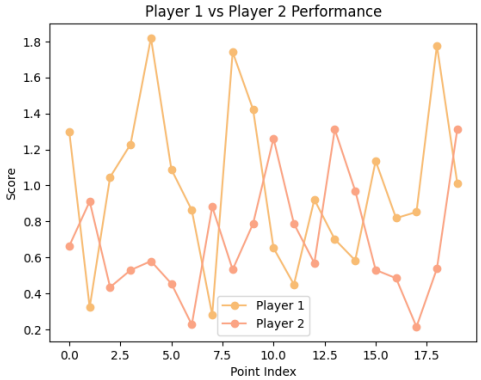


Figure.7 Comparison line chart of momentum for both players at a specific time

3. Assess and Predict Momentum Fluctuations

3.1. Model Construction

In the last section, we have constructed a predictive model based on a Random Forest model. We have formulated and implemented quantifiable momentum formulas and functions. Now, we are furthering our research in this context. We define a new function called 'momentum_diff,' which represents the difference in momentum at different points. Subsequently, we trans-form momentum changes into categorical variables: 1 for increasing, -1 for decreasing, and 0 for unchanged. Then, we define a function called 'momentum_change' to record these transformed categorical variables. If 'momentum_change' starts increasing, it indicates that we are gaining momentum, whereas if it starts decreasing, it signifies the opponent is gaining momentum. If there is little to no change over a period, it suggests a balanced game.

3.1.1 Feature Selection and Correlation Calculation

To filter out the most relevant factors, we employed a Firefly Algorithm-optimized Random Forest model (FA-RF), which combines the powerful data processing capabilities of Random Forest (RF) with the optimization search capabilities of the Firefly Algorithm (FA) [6].

We used the Firefly Algorithm to optimize the feature selection process, identifying the feature variables that have the most significant impact on match outcomes, such as player scores, running distance, and more.

Additionally, we utilized the Firefly Algorithm to optimize the parameters of the Random Forest model, including the number of decision trees and the maximum tree depth. With the optimized parameter settings, we trained and learned from the tennis match results using the Random Forest model.

In the context of tennis matches, each firefly represents a set of predictive factors for the tennis match. The brightness of a firefly maps to the accuracy or predictive performance of the predictive

model based on specific factors. The attraction between fireflies maps to the relative utility or advantage of different factors. More effective factors will "attract" other factors towards them. Here are the key formulas used:

$$\beta(r) = \beta_0 e^{-\gamma r^2} \tag{9}$$

$$x_i^{t+1} = x_i^{t-1} + \beta(r)\left(x_j^t - x_i^t\right) + \alpha(rand - 0.5) \tag{10}$$

Where Formula (7) is the attraction calculation formula used to assess the contribution of different factor combinations to the accuracy of predicting outcomes. Formula (8) is the position update formula that guides how to adjust factor combinations to gradually find the optimal factor configuration, i.e., the one that maximizes predictive accuracy.

Through the FA-RF model, we can utilize the Firefly Algorithm to optimize the selection of factors for the tennis match result prediction model, thereby enhancing the accuracy and reliability of predictions. This process involves continuous experimentation and adjustment of different factor combinations until the best combination is found. We use the model to calculate the confusion matrix, and obtain a heatmap illustrating the importance relationship between features and 'momentum change.' To provide a more intuitive view of the importance of individual feature values, we have generated rectangular plots as shown in Fig.8 and Fig.9.

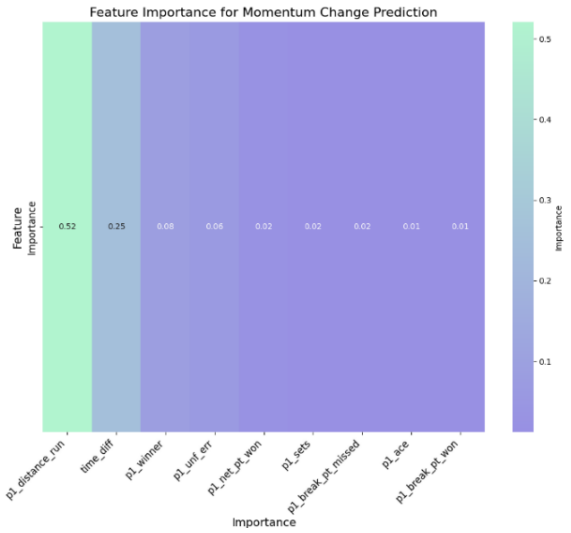


Figure.8 Importance heatmap

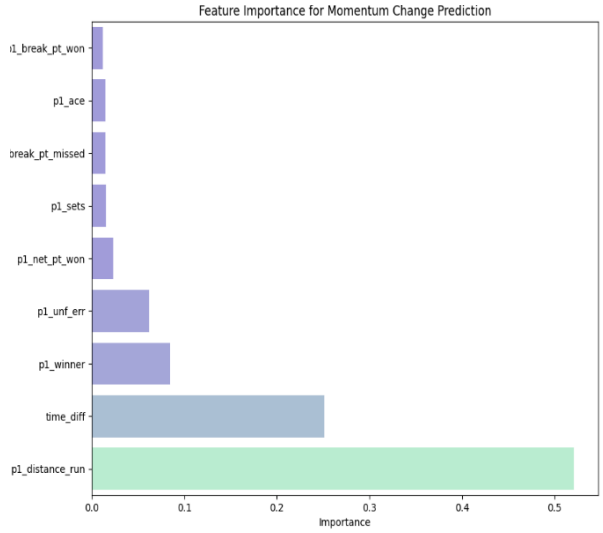


Figure.9 Importance rectangular plot

3.1.2 Model Validation and Data Prediction

According to model, we input all the data and divide it into five groups, with four groups used for training and one group for testing. Some key metrics of model are shown in Table 2 below next page.

In addition, we conducted cross-validation to assess the robustness of the statistical analysis and avoid overfitting of the data. Cross-validated scores are a list containing multiple scores, with each score corresponding to one iteration during the cross-validation process. In this case, our model was trained and validated five times (5-fold cross-validation), each time using a different subset of data. The accuracy scores achieved in each of these iterations were 0.7390, 0.7362, 0.7356, 0.7232, and 0.7497.

Table 2. The relevant data of the model's prediction results

	Precision	Recall	F1	Support
Decrease	0.7397	0.7487	0.7442	816
Increase	0.7631	0.7595	0.7613	855
Accuracy	0.7507	0.7507	0.7507	/
Macro Avg	0.7342	0.7361	0.7352	1685
Weighted Avg	0.7481	0.7507	0.7493	1685

The Average cross-validation score represents the mean of these five iterations' accuracy scores, providing an overall performance metric obtained average cross-validation score is 0.74, indicating that the model's average accuracy is 74%, which is quite good.

Subsequently, we use the Standard deviation of cross-validation scores to indicate the standard deviation of the cross-validation scores, which provides information about the consistency of the model's performance. A smaller standard deviation implies more stable model performance. The standard deviation we obtained is 0.01, indicating that the model's performance is relatively consistent across different data subsets, with minimal fluctuation.

In summary, the above values provide methods to measure the model's performance on different data subsets and offer indicators of the stability and reliability of the model's performance. Our final model achieved an accuracy of 74%, which demonstrates good predictive results.

3.2. Predicted results visualization

After completing the model building mentioned above, we made predictions on the da-ta and compared it with our actual data. Due to the large volume of data overall, for ease of observation and to demonstrate the predictive results, we have extracted a subset of data for display. The overall prediction performance is shown in Fig.10 below:

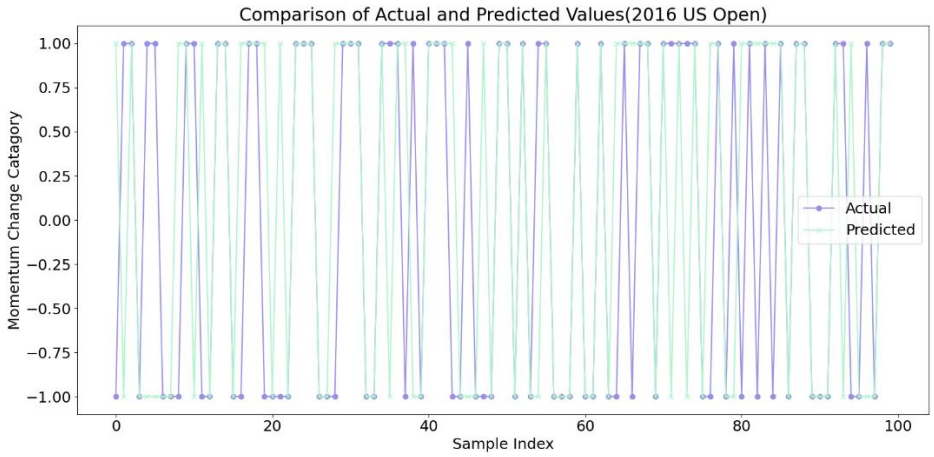


Figure.10 Line chart comparing actual values with predicted values

In Fig.10, we can intuitively see that our model has performed well overall, with minimal deviations, achieving the expected results.

4. Summary

The Markov chain model was used to successfully capture the match point and the game flow, and the real-time observation of the change of the player's win rate during the game was realized. The Stochastic Forest Model (FA-RF), optimized based on factor analysis, enables the evaluation of a player's performance at a given moment and provides a deeper understanding of game dynamics and fluctuations through the visualization of momentum. The correlation between momentum and the outcome of a match is studied.

The change of momentum is used to predict the matching trend, and the accuracy of the model is improved by training with the Random Forest algorithm (FA-RF). Research has shown that a player's distance of travel, progress of the match, consecutive points and un-forced errors significantly affect momentum and, in turn, the trend of the match. The accuracy of the model is about 74% on average through different data sets, which proves the high reliability of the model.

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